Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations

Anonymous ECCV 2026 Submission

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Abstract. Multimodal Emotion Recognition in Conversations (MERC) aims to identify emotions in each utterance using visual, audio, and linguistic cues, with applications in affective computing and human-computer interaction. Among the various approaches proposed to solve this problem, graph based approaches have emerged as some of the most promising ones but they often underutilize the visual modality due to noisy pixel level features. Traditional visual encoders focus on geometric and texture patterns, which often fail to capture facial expressions and behavioral cues, resulting in weak cross-modal alignment. To address this issue, we propose a Vision Language Model (VLM) based approach that encodes facial expressions and behavioral cues from the visual modality into semantically meaningful representations. Each utterance's video is translated into a descriptive textual representation using structured VLM prompts capturing expressions, gestures, body language and scenario understanding. These semantic visual features replace traditional low level visual inputs in graph based MERC frameworks, reducing noise and allowing the model to focus on meaningful visual information. This work presents VLM based semantic visual feature extraction as a viable replacement for conventional visual feature extraction pipelines in graph based MERC frameworks.

Keywords: Multimodal Emotion Recognition in Conversations · Vision Language Models · Graph Neural Networks

1 Introduction

The proper interpretation and communication of emotions are crucial, as they significantly influence interactions, conversations, and decision-making processes. In an AI-driven future, it is essential that emerging systems can recognize emotions at or above human ability. Understanding emotions in conversations remains a fundamental challenge in human–computer interaction and affective computing [5].

Multimodal Emotion Recognition in Conversations (MERC) aims to identify emotions in each utterance by leveraging textual, acoustic, and visual signals

[28,36]. Unlike single-utterance emotion recognition, MERC considers contextual dependencies across speakers and time, making it a more realistic and complex problem [10,24]. Approaches range from early fusion techniques and transformer-based models to more structured, context-aware graph based architectures [10, 14, 32, 34, 35].

Graph based architectures, inspired by Graph Convolutional Networks (GCNs) [17], are particularly effective for MERC [10, 14]. By modeling conversations as graphs with utterances as nodes and relational dependencies as edges, these models capture both intra and inter-speaker interactions, achieving competitive performance [14].

Despite these advances, visual features remain underutilized in many graph based MERC frameworks [14, 19]. Traditional visual encoders such as CNNs [13] or Vision Transformers (ViTs) [7] extract low level spatial or patch based representations, which often fail to capture subtle behavioral cues and body language necessary for emotion understanding [21]. Aggregating frame level embeddings into utterance level features further collapses temporal dynamics, averaging out expressive shifts within an utterance [34], weakening cross-modal alignment compared to textual or acoustic modalities [6, 31].

To address this limitation, we propose a Vision Language Model (VLM)-based approach that encodes visual content as semantically meaningful descriptions. Using VidEmo-3B (based on Qwen2.5-VL-3B [1]), each utterance-level video is translated into a structured textual narrative capturing facial expressions, head movements, and body language. These high-level semantic features preserve intra-utterance dynamics and align better with linguistic and acoustic modalities [29]. Replacing conventional low level visual embeddings with these features in graph based MERC architectures reduces noise, improves cross-modal consistency, and encourages the model to leverage meaningful visual cues.

In summary, this work makes the following contributions:

1. We identify and analyze the underutilization of the visual modality in graph based MERC due to low level, noisy feature representations.
2. We propose a VLM-based semantic visual feature extraction approach that converts raw video into structured textual descriptions capturing emotion-relevant cues.
3. We integrate these semantic visual features into graph based MERC architectures, promoting stronger cross-modal alignment.

By bridging the gap between visual perception and semantic understanding, our approach offers a new direction for enhancing the role of visual information in multimodal conversational emotion recognition.

2 Related Works

2.1 Emotion Recognition in Conversations and Multimodal Extensions

Emotion Recognition in Conversations (ERC) aims to identify the emotional state of each utterance by leveraging dialogue context and speaker interac-

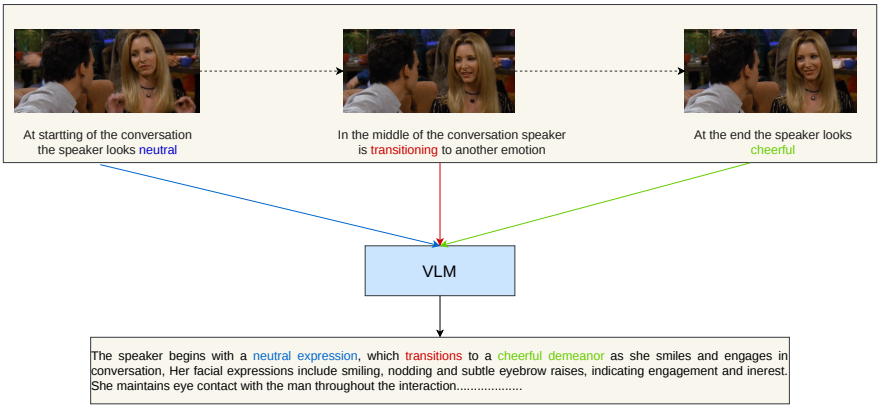

Fig. 1: Overview of the proposed semantic visual preprocessing pipeline.

tions [32]. Early sequence models using RNNs or GRUs capture short term dependencies and speaker states [11, 24], but often struggle with long range interactions.

Multimodal ERC (MERC) incorporates text, acoustic, and visual cues to better reflect human emotion expression [28, 36]. Text provides lexical and semantic content, acoustic features convey speech patterns [31], and visual data encodes facial expressions and gestures [21]. MERC introduces additional challenges, including modality alignment, noise handling, and effective cross-modal fusion [32].

2.2 Approaches to MERC

To address these challenges, several modeling strategies have been proposed.

Sequence and Context Based Methods. Sequence models capture temporal dependencies and speaker states in conversations. DialogueRNN [24] uses recurrent structures to leverage dialog history for emotion prediction but has limited ability to model complex speaker or cross-modal interactions.

Fusion Based Methods. Fusion approaches integrate multimodal information using different strategies. Early and late fusion methods combine modality features either at the input level or at the decision stage and are widely used baselines in multimodal sentiment and emotion analysis [25–27, 37]. More advanced approaches employ attention mechanisms to model interactions between modalities. Multimodal Transformer architectures use cross modal attention to align textual, acoustic, and visual representations [34], while MISA learns modality invariant and modality specific representations for improved multimodal integration [12]. Additionally, models such as DialogueRNN and related methods

incorporate mechanisms to track speaker states and emotion transitions within conversations [11, 24].

Although fusion methods improve multimodal interaction, they often do not explicitly represent conversational structure or relational dependencies between speakers.

2.3 Graph Based MERC Approaches

Graph based methods are powerful for MERC as they encode conversational structure and relational dependencies [14, 19]. Utterances are nodes and edges represent speaker interactions, temporal continuity, and cross-modal links [17]. DialogueGCN [10] introduced graph reasoning for ERC, modeling intra- and inter-speaker relations and outperforming sequential models. Extensions like MMGCN [14] and hypergraph models capture multimodal interactions, while **Joyful** [18] combines graph fusion with contrastive learning to improve multimodal representations and address issues like oversmoothing.

GraphSmile [19] represents a state-of-the-art graph-centric MERC architecture introducing two key components: *Graph Structure Fusion (GSF)* [19], which constructs bipartite dialogue graphs [14] between every pair of modalities and alternates information propagation across these graphs to prevent any single structural path from dominating; and *Sentiment Dynamics Perception (SDP)* [19], an auxiliary task that explicitly predicts whether sentiment shifts occur between consecutive utterances to guide the model toward emotional transitions. Despite its strong performance, GraphSmile, like most existing graph based MERC methods depends on conventional visual encoders (*e.g.*, CNN backbones [13] or handcrafted facial features [2]) that produce static, low level visual embeddings. These embeddings are typically mean-pooled over frames, collapsing intra utterance temporal dynamics and limiting the visual modality's contribution to emotional reasoning. Graph based approaches excel at structurally modeling dialogues and multimodal interactions, but their effectiveness is often constrained by the quality of unimodal representations feeding into the graph.

2.4 Vision Language and LLM Based Approaches for Emotion Recognition

To overcome limitations of low level visual features, recent research has explored Vision Language Models (VLMs) and multimodal Large Language Models (LLMs) for emotion recognition tasks. VLMs pretrained on large image/video text datasets [29] produce semantically meaningful representations of visual content: they generate textual or semantic descriptions from visual inputs, capturing behaviors, expressions, and dynamics that are difficult for CNN [13] or ViT [7] features to represent, and these descriptions are inherently aligned with language modalities [6], facilitating cross modal fusion and multimodal reasoning.

Recent works demonstrate that VLMs can capture emotionally relevant visual cues. Models such as VidEmo-3B [38] leverage video–language pretraining

to describe visual affective cues over time, producing semantically grounded visual narratives. Task specific adaptations like Emotion-Qwen [15] build hybrid frameworks that integrate emotion reasoning into general vision–language understanding by combining specialized emotion experts with general VL backbones. Evaluation studies of general VLMs show that, with appropriate prompting or fine tuning, vision–language models can recognize nuanced emotional expressions and behaviors that traditional encoders miss.

Multimodal LLM integrations extend this idea further. Works like BeMERC [9] propose behavior aware MLLM frameworks for MERC, using multimodal LLMs tuned to behavior features (*e.g.*, facial motion, body language) to directly predict conversational emotions, showing that LLMs can leverage both visual and linguistic context effectively. Prompt based or zero shot methods demonstrate that general large vision language systems such as CLIP [29] or Qwen2.5-VL [1] can be prompted to interpret emotional cues in video or images without explicit retraining, offering flexible emotion inference.

These VLM/LLM approaches address two key limitations in traditional MERC: visual modality underutilization, by generating high level semantic visual descriptions rather than low level features [21]; and cross modal alignment challenges, by inherently aligning visual and language modalities through shared semantic outputs [29]. Integrating semantic VLM representations into structured MERC frameworks, especially graph based methods, emerges as an important frontier for multimodal emotion recognition.

2.5 Positioning of Our Work

Our work builds upon the strengths of graph based MERC frameworks , particularly GraphSmile [19], while addressing the underutilization of visual information. Instead of using static, low level visual embeddings [2, 13], we introduce a VLM based semantic visual preprocessing stage [1, 38] that converts raw video segments into temporally structured textual narratives capturing facial expressions, gaze shifts, and body language evolution within an utterance.

These semantic visual descriptions are encoded into dense vectors [6] and integrated into the Graph Structure Fusion (GSF) pipeline [19]. This design preserves intra utterance temporal dynamics, improves cross modal alignment, reduces reliance on noisy visual appearance features, and enhances the model's ability to leverage visual cues in relational reasoning across conversation turns. By combining high level semantic visual reasoning from VLMs with structured graph propagation, our approach advances MERC beyond existing graph based and VLM based methods.

3 Methodology

The proposed framework builds upon the GraphSmile architecture [19] by introducing a novel approach that converts raw visual inputs into meaningful semantic representations before multimodal fusion.

While prior graph based Multimodal Emotion Recognition in Conversation (MERC) systems relied on conventional visual encoders (such as DenseNet for extracting abstract spatial features [16] and OpenFace for tracking deterministic facial mechanics like Action Units [2]), our methodology moves beyond these restricted appearance descriptors. By leveraging a temporally aware Vision Language Model (VLM) [1,29], we encode the rich, semantically grounded behavioral signals present in human speech.

The overall architecture comprises three main stages:

1. The independent extraction of utterance level features across textual, acoustic, and visual modalities.
2. The construction of multimodal dialogue graphs for alternating propagation via the Graph Structure Fusion (GSF) module [19].
3. Multitask classification incorporating a Sentiment Dynamics Perception (SDP) module [19].

3.1 Utterance Level Feature Extraction

Each modality is processed independently at the utterance level before being integrated into the multimodal graph. In contrast to the visual channel, the textual and acoustic modality remains structurally unchanged relative to the GraphSmile baseline [19], preserving a controlled point of comparison.

Textual Feature Extraction. For the textual modality, utterance transcripts are encoded uniformly across both datasets using a pretrained transformer language model, specifically RoBERTa [22]. The `<s>` (classification) token representation of each transcript forms the textual feature vector, X^t, capturing lexical and syntactic sentiment signals.

Acoustic Feature Extraction. The acoustic features, X^a, capture paralinguistic speech characteristics including fundamental frequency (F0), energy, spectral properties, and temporal statistical functionals that reflect prosody and pitch dynamics. For both the IEMOCAP (6-way) [4] and MELD [28] datasets, these utterance level features are extracted using the openSMILE toolkit with the IS10 configuration [8].

Semantic Visual Feature Extraction. Traditional visual encoders often fail to fully harness conversational cues because they are restricted to extracting context blind spatial features or isolated facial mechanics, creating a semantic gap between raw physical movements and complex emotional states. The core methodological contribution addresses this by redefining the visual feature extraction process through a three stage semantic preprocessing pipeline.

Stage 1: Dataset Specific Video Preprocessing. To ensure the VLM accurately analyzes the correct subject, we apply tailored preprocessing strategies

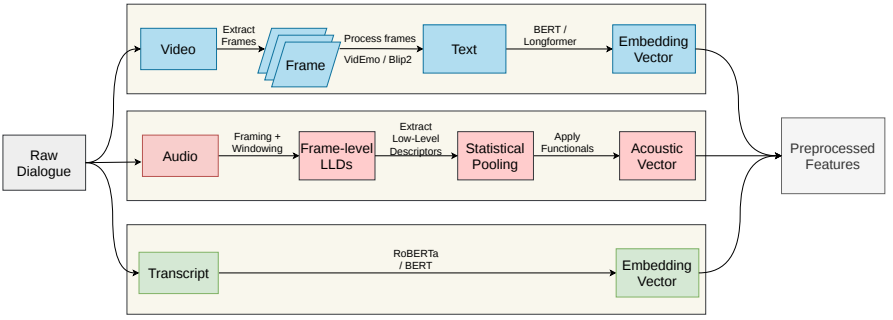

Fig. 2: The multimodal preprocessing pipeline for extracting utterance-level textual, acoustic, and visual features.

based on the native format of each dataset. The IEMOCAP dataset [4], which consists of continuous, multi speaker dialogue recordings, undergoes two explicit preprocessing steps. First, the sessions are temporally segmented into discrete, utterance level clips using provided timestamp annotations; second, each frame is spatially cropped to physically isolate the active speaker and eliminate visual interference. In contrast, the MELD dataset [28] inherently provides presegmented utterance level clips, bypassing the need for physical cropping. Instead, spatial isolation for MELD is achieved entirely through zero shot prompting, wherein the VLM is explicitly instructed to autonomously identify and focus exclusively on the active speaker within the uncropped, multi person frame.

Stage 2: VLM Temporal Analysis. Each utterance clip is processed by VidEmo-3B [38] (built on the Qwen2.5-VL-3B backbone [1]). The model is driven by a highly structured prompt instructing it to focus exclusively on the active speaker and ignore any other visible participants. The VLM maps the speaker's emotional evolution by dividing the clip into a temporal progression (beginning, middle, and end). For each segment, the model grounds its analysis in specific visible signals across four dimensions:

- **Facial Expressions:** Specific muscle movements of the brow, lips, eyes, jaw, and cheeks.
- **Head & Gaze:** Tilts, nods, shakes, and eye direction.
- **Body Language:** Posture shifts, gestures, and physical tension or relaxation.
- **Instantaneous Emotion:** The most likely emotional state, explicitly supported by the aforementioned visual cues.

The temporal analysis concludes with a one sentence summary of the overall emotional arc (*e.g.*, *starts neutral* $\rightarrow$ *builds frustration* $\rightarrow$ *brief smile at end*). By enforcing that every observation is grounded in a specific visible signal from the speaker, this stage generates a semantically dense, temporally aware description of affective behavior, bypassing the noise of traditional frame averaging encoders.

In a separate experimental condition on the MELD dataset, BLIP-2 [20] is employed as an alternative visual analysis model within this stage. Rather than processing the full video clip, three representative frames are sampled at fixed temporal positions corresponding to the beginning, middle, and end of each utterance. Each frame is individually processed by BLIP-2 with a prompt focused exclusively on the emotional state of the speaker in that moment, and the resulting per-frame descriptions are concatenated into a single utterance-level textual representation. This frame-level approach produces more targeted emotional descriptions, reducing scene-level noise that is particularly prevalent in the multi-party MELD setting.

Stage 3: Text Encoding of VLM Output. To bridge the symbolic output of the VLM with the continuous feature space expected by the graph network, the free text description is encoded using a pretrained transformer language model. Two encoders are evaluated in separate experimental conditions: BERT [6], which truncates inputs to 512 tokens and extracts the `[CLS]` token embedding; and Longformer [3] , which processes the full description without truncation. Both produce a 768-dimensional dense vector that serves as the new visual modality feature, X^v. Encoding visual behavior in the same semantic space as the transcript fosters superior cross modal alignment within the GSF module [19].

3.2 Graph Structure Fusion (GSF) Module

The Graph Structure Fusion (GSF) module processes dialogue graphs layer by layer, alternating between sequential edges for temporal flow and speaker edges for personal consistency. This back-and-forth approach prevents any single relationship from dominating and effectively separates a speaker's internal emotional shifts from the conversation's external influences.

Multimodal Dialogue Graphs To understand how utterances relate to each other across different sensory inputs, we build three bipartite dialogue graphs [14, 19]. Each graph connects two specific modalities. For example, the textual visual graph is defined as $\mathcal{G}^{tv} = (\mathcal{N}^t, \mathcal{N}^v, \mathcal{E}^{t\leftarrow v}, \mathcal{E}^{v\leftarrow t})$. Here, $\mathcal{N}^t$ and $\mathcal{N}^v$ stand for the sets of textual and visual nodes. The edge sets $\mathcal{E}^{t\leftarrow v}$ and $\mathcal{E}^{v\leftarrow t}$ represent the connections directed from visual to textual nodes, and vice versa.

- **Nodes:** Every utterance u_i is represented by a textual node n_i^t and a visual node n_i^v.
- **Edges:** We only draw edges between nodes of opposite modalities. To keep the graph from becoming too dense, we use a sliding window approach. A textual node n_i^t connects to visual nodes from the past P steps (i.e., $n_{i-P}^v, \ldots, n_{i-1}^v$) and the future F steps (i.e., $n_{i+1}^v, \ldots, n_{i+F}^v$).
- **Edge Weights:** These connections are directed to show the specific flow of information. We assign trainable weights to these edges, which start at a value of 1 and are fine tuned as the model trains.

– **Adjacency Matrix:** All of these node and edge relationships are stored in an adjacency matrix. This matrix is the core structural input for the graph neural network.

Alternate Graph Propagation Graph Neural Networks (GNNs) [17, 30] are highly effective at uncovering both cross modal and intra modal relationships. In our design, each dialogue graph passes information through multiple simplified convolutional layers. It alternates between extracting inter modal and intra modal features layer by layer, calculated as:

$$X^{(l)} = AX^{(l-1)}\Theta^{(l)}, \quad l = 1, 2, \ldots, L$$

Here, A is the dialogue graph's adjacency matrix, $X^{(l)}$ is the output feature at layer l, and $\Theta^{(l)}$ is the trainable weight matrix for that layer. For the textual-visual graph, the starting input $X^{tv,(0)}$ simply combines the raw textual features $[x_1^t, \ldots, x_M^t]$ and visual features $[x_1^v, \ldots, x_M^v]$.

Next, we use residual connections [13] to sum the feature representations from every layer:

$$X = X^{(0)} + \mathrm{FC}^{(1)}(X^{(1)}) + \cdots + \mathrm{FC}^{(L)}(X^{(L)})$$

The fully connected layer $\mathrm{FC}(\cdot)$ includes a linear transformation, a LeakyReLU [23] activation, and dropout [33]:

$$\mathrm{FC}(X) = \mathrm{Dropout}(\mathrm{LeakyReLU}(X\Theta_f + b_f))$$

This setup serves two purposes: it lets the network dynamically choose which layer's information is most useful, and it boosts the overall expressive power of the GSF module.

Multimodal Integration After processing the textual-visual ($\mathcal{G}^{tv}$), textual-acoustic ($\mathcal{G}^{ta}$), and visual-acoustic ($\mathcal{G}^{va}$) graphs through their own separate GSF modules, we get three distinct outputs: X^{tv}, X^{ta}, and X^{va}. We add these together to create a final, unified feature representation:

$$\begin{aligned} H = {} & \sigma\big((X^{t\leftarrow v})\Theta_h\big) + \sigma\big((X^{v\leftarrow t})\Theta_h\big) \\ & + \sigma\big((X^{t\leftarrow a})\Theta_h\big) + \sigma\big((X^{a\leftarrow t})\Theta_h\big) \\ & + \sigma((X^{v\leftarrow a})\Theta_h) + \sigma((X^{a\leftarrow v})\Theta_h) \end{aligned} \tag{1}$$

In this equation, σ is the activation function (like LeakyReLU) and Θ_h is a shared learning parameter. This combined matrix H effectively captures all the complex, multi sensory emotional signals in the conversation.

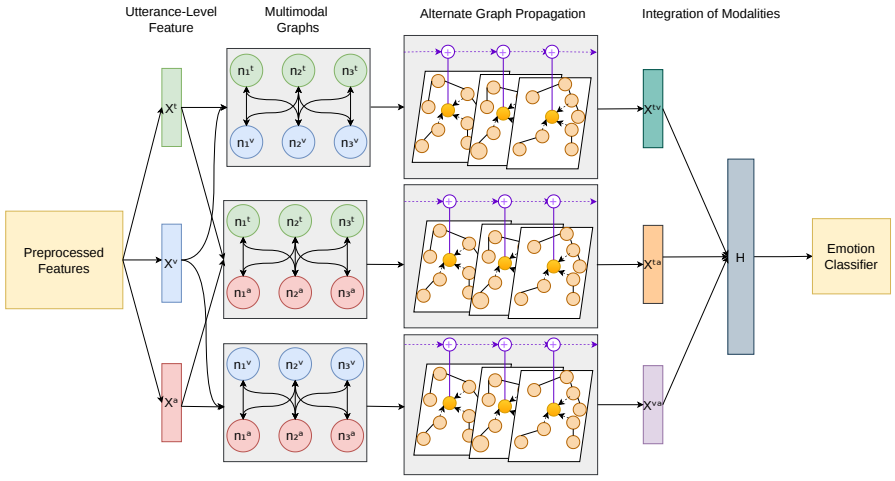

Fig. 3: Multimodal dialogue graph structure used in GraphSmile [19], in which utterance nodes from different modality pairs are connected via bipartite graphs and propagated through the Graph Structure Fusion (GSF) module. Adapted from [19].

3.3 Emotion Classification

After the graph propagation and integration phase, the resulting feature matrix H is passed into a multitask classification head. For the primary task of emotion classification, H goes through a linear classifier and a softmax function to predict the exact emotional category. This is formulated as:

$$\hat{Y} = \text{softmax}(H\Theta_y + b_y)$$

$$\hat{e}_i = \text{argmax}(\hat{y}_i[\tau])$$

Here, Θ_y and b_y are the learned weights and biases. The vector $\hat{y}_i$ holds the emotion probabilities for utterance u_i, and $\hat{e}_i$ is the final predicted emotion label. We optimize the network using categorical cross-entropy loss:

$$\mathcal{L}_e = -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{C_e} y_{i,j}\log\hat{y}_{i,j}$$

where M is the total number of utterances, C_e is the number of emotion classes, $y_{i,j}$ is the true label, and $\hat{y}_{i,j}$ is our model's prediction.

3.4 Sentiment Dynamics Perception (SDP) Module

Existing dialogue models often focus on static context, missing crucial inter utterance sentiment changes. To address this, we introduce the Sentiment Dynamics Perception (SDP) module [19] as a vital auxiliary task. The SDP module

frames conversation dynamics as a pairwise prediction problem, tracking whether sentiment shifts or remains constant between consecutive utterances. By jointly optimizing this auxiliary shift loss with the main emotion classification loss, the network is strictly guided to learn the dialogue's emotional momentum. This joint supervision improves class separation (pulling similar sentiments closer and pushing differing ones apart), while perfectly aligning with the temporal descriptions created during the VLM preprocessing stage.

Shift Label We define a sentiment shift label $o_{(i,j)}$ for any pair of utterances u_i and u_j. If the sentiment changes between the two utterances, the label is 1; if the sentiment remains the same, the label is 0:

$$o_{(i,j)} = \begin{cases} 1, & \text{if } s_i \neq s_j, \\ 0, & \text{if } s_i = s_j. \end{cases}$$

where s_i denotes the ground truth sentiment state of utterance u_i.

Shift Classification We construct pairwise shift features T from the fused representations and pass them through a softmax classifier to predict the shift probability $\hat{Z}$:

$$\hat{Z} = \text{softmax}(T\Theta_z + b_z)$$

where Θ_z and b_z are learnable parameters. The module is optimized using a standard cross-entropy loss function, denoted as $\mathcal{L}_o$.

3.5 Sentiment Classification

To extend our model to the Multimodal Sentiment Analysis in Conversation (MSAC) task, we add a sentiment classifier. The fused feature matrix H is processed through a linear layer and a softmax activation:

$$\hat{W} = \text{softmax}(H\Theta_w + b_w)$$

Here, Θ_w and b_w are learnable parameters, and $\hat{W}$ contains the predicted sentiment probabilities. This task is supervised by a sentiment classification cross-entropy loss, denoted as $\mathcal{L}_s$.

3.6 Total Training Objective

To simultaneously extract both emotional and sentimental cues, the primary emotion recognition task is jointly learned alongside the sentiment and shift classification tasks. The overall network parameters are optimized using a combined loss function:

$$\mathcal{L}_{total} = \mathcal{L}_e + \lambda_s \mathcal{L}_s + \lambda_o \mathcal{L}_o + \beta ||\Theta_{total}||$$

where $\mathcal{L}_e$ is the primary emotion classification loss, λ_s and λ_o are trade-off hyperparameters balancing the auxiliary tasks, β is the L_2 regularization weight, and Θ_{total} represents all trainable parameters in the model.

4 Experimental Details

4.1 Datasets Overview

IEMOCAP [4]**.** The dataset consists of continuous, dyadic conversational sessions recorded with a dual-camera setup, presenting both actors side-by-side in a single unsegmented video frame. The accompanying audio is recorded in stereo, capturing each speaker on a dedicated independent microphone channel.

MELD [28]**.** This dataset is constructed from episodes of the television series *Friends*, representing a multi-party conversational environment. The video data is inherently provided as pre-segmented, utterance-level clips.

Table 1: Dataset statistics for IEMOCAP and MELD.

Dataset	Dialogues	Utterances	Classes	Train / Val / Test
IEMOCAP	151	7,433	6	5,810 / —/ 1,623
MELD	1,432	13,708	7	9,989 / 1,108 / 2,611

4.2 Utterance-Level Video Preparation

IEMOCAP.

- Segment each conversation into utterance level clips using timestamps.
- Crop frames to isolate the active speaker.
- Retain only the speaker's audio channel from the stereo stream.
- Output speaker focused utterance level video clips.

MELD.

- Pre-segmented utterance level clips; no cropping needed.
- Each clip corresponds to a single speaker utterance.
- Speaker focus is handled during prompting.

4.3 Semantic Visual Feature Extraction

Video-to-Text Generation. Vision Language Models (VLMs) are utilized to generate descriptive text for each utterance. The VidEmo-3B model [38], built upon the Qwen2.5-VL backbone [1], is applied to both datasets.

Frame Sampling. IEMOCAP videos are sampled at a fixed rate of 1 frame per second (FPS). MELD videos use an adaptive strategy of 1 FPS for clips under 15 seconds, capped at a maximum of 16 frames for longer clips.

Prompt Structure. The prompt requests a temporal progression (beginning, middle, and end) detailing facial expressions, head and gaze, body language, and instantaneous emotion, concluding with a one-sentence emotional arc summary.

For MELD, zero-shot instructions are included to focus exclusively on the active speaker.

Batching and Computation. For IEMOCAP, inference was executed with a batch size of 4, requiring over 7 hours. For MELD, inference was executed with a batch size of 2, requiring over 10 hours.

Text-to-Vector Encoding. The generated descriptions are encoded using two models evaluated in separate experimental conditions:

- **Longformer** [3] : inputs are processed without truncation; a 768-dimensional embedding is extracted from the `[CLS]` token.
- **BERT** [6] : inputs are truncated to a maximum of 512 tokens; a 768-dimensional embedding is extracted from the `[CLS]` token.

4.4 Implementation Details

Hardware and Environment. All experiments are conducted on an NVIDIA GeForce RTX 3060 (12 GB VRAM) using a Conda environment with PyTorch 2.8.0+cu126.

Table 2: Hyperparameter settings for IEMOCAP and MELD.

Hyperparameter	IEMOCAP	MELD
Epochs	120	50
Batch Size	16	16
Learning Rate	1×10^{-4}	7×10^{-5}
Hidden Dimension	512	384
Sliding Window	17	3
Shift Window	19	3
Heterogeneous Graph Layers	7	4
Dropout Rate	0.2	0.2
Loss Weights	1.0, 1.0, 0.7	1.0, 0.5, 0.2

4.5 Integration into Graph Based MERC Framework

Feature Substitution and Dimensionality. The 768-dimensional visual embeddings generated via the VLM and subsequent text encoding are substituted for the original 342-dimensional handcrafted visual descriptors in the dataset manifest. A linear projection layer is implemented at the input of the visual processing stream to map the 768-dimensional vectors into the hidden dimension space of the graph architecture. Textual and acoustic features remain identical to the GraphSmile baseline [19], with dimensions of 1024 (RoBERTa) and 1582 (openSMILE IS10) for IEMOCAP, and 1024 and 300 respectively for MELD.

Evaluation Metrics. Accuracy (ACC) and Weighted F1-score (WF1) serve as the primary evaluation metrics for overall model performance. The F1-score is

Table 3: Main Results for the MERC Task on the IEMOCAP Dataset.

Method	VLM Model	Text Encoder	IEMOCAP-6							
			Happy	Sad	Neutral	Angry	Excited	Frustrated	ACC	WF1
COGMEN†	–	–	51.90	81.70	68.60	66.00	75.30	58.20	68.20	67.60
Joyful†	–	–	60.94	84.42	68.24	69.95	73.54	67.55	70.55	71.03
DER-GCN†	–	–	58.80	79.80	61.50	72.10	73.30	67.80	69.70	69.40
DialogueCRN	–	–	53.85	82.66	71.03	62.33	77.64	58.81	68.70	68.82
MMGCN	–	–	47.10	81.91	66.44	63.51	76.17	59.06	67.10	66.81
MM-DFN	–	–	43.36	83.23	70.03	70.19	73.11	64.01	69.44	68.83
SACL-LSTM	–	–	51.30	82.25	71.39	67.78	75.26	66.94	70.55	70.60
M3Net	–	–	60.93	78.84	70.14	68.06	77.11	67.42	70.92	71.07
GraphSmile‡	–	–	59.31	82.47	69.68	67.53	78.76	66.06	71.23	71.29
GraphSmile+VLM‡	Qwen-VL-3B	Longformer	54.11	82.82	70.23	70.72	79.81	65.87	71.53	71.49
GraphSmile+VLM‡	Qwen-VL-3B	BERT	61.49	83.43	69.69	67.33	77.12	67.51	71.53	71.65

further utilized to evaluate classification performance for each individual emotion category.

Training and Optimization Strategy. The model follows the multi-task optimization strategy of GraphSmile [19], jointly learning the primary emotion recognition task alongside sentiment and sentiment-shift classification. All structural components of the Graph Structure Fusion (GSF) module and the Sentiment Dynamics Perception (SDP) module are maintained as per the original implementation to ensure a controlled comparison of visual feature effectiveness.

5 Results and Observations

5.1 Results Comparison

Tables 3 and 4 summarize the performance of the proposed semantic visual representation integrated with GraphSmile on the IEMOCAP and MELD datasets. The evaluation includes overall accuracy (ACC), weighted F1 score (WF1), and per-class F1 scores for each emotion category.

On the IEMOCAP dataset, GraphSmile with BERT-based encoding of VLM-generated descriptions achieved the highest performance, with a weighted F1 score of 71.65, slightly improving over the baseline GraphSmile (WF1 71.29). Longformer-based encoding also performed comparably (WF1 71.49), but did not surpass BERT. For individual emotions, BERT-based VLM representations improved recognition for Happy (61.49) and Frustrated (67.51) categories.

On the MELD dataset, the improvements were more modest. GraphSmile integrated with BLIP-2 [20] and BERT achieved the best performance among the VLM variants, with ACC 67.47 and WF1 66.35. Qwen-VL-3B with BERT also slightly outperformed Longformer-based encoding (ACC 67.32 vs. 67.13; WF1 66.12 vs. 65.97). Overall, performance remained close to the GraphSmile baseline (ACC 67.66, WF1 66.67). Individual emotion recognition showed minor variations across Neutral, Joy, Sadness, and other categories.

Table 4: Main Results for the MERC Task on the MELD Dataset.

Method	VLM Model	Text Encoder	MELD								
			Neutral	Surprise	Fear	Sadness	Joy	Disgust	Anger	ACC	WF1
Joyful†	–	–	76.80	51.91	–	41.78	56.89	–	50.71	62.53	61.77
DER-GCN†	–	–	80.60	51.00	10.40	41.50	64.30	10.30	57.40	66.80	66.10
DialogueCRN	–	–	76.15	56.72	18.67	38.29	63.21	27.69	50.67	62.38	63.32
MMGCN	–	–	78.62	57.78	3.77	40.35	63.60	12.20	53.68	66.02	64.55
MM-DFN	–	–	79.84	58.43	15.79	31.65	64.01	28.04	53.60	67.05	65.21
SACL-LSTM	–	–	77.42	58.50	20.41	39.58	62.76	34.71	52.08	64.52	64.55
M3Net	–	–	79.14	59.54	13.33	42.86	65.05	21.69	53.54	66.59	65.83
GraphSmile‡	–	–	80.32	58.96	18.18	42.46	64.99	32.43	53.59	67.66	66.67
GraphSmile+VLM‡	Qwen-VL-3B	Longformer	79.82	57.56	16.44	41.95	63.94	31.78	53.19	67.13	65.97
GraphSmile+VLM‡	Qwen-VL-3B	BERT	79.82	57.73	19.18	42.17	63.94	32.08	53.61	67.32	66.12
GraphSmile+VLM‡	BLIP2-2.7B	BERT	79.78	58.30	14.29	44.20	64.56	30.77	54.00	67.47	66.35

The marker † indicates the results (%) as replicated by the GraphSmile paper, and the marker ‡ indicates the results (%) replicated by us and no marker indicates results taken from original papers.

5.2 Observations

The results demonstrate that semantic visual representations can effectively replace traditional handcrafted visual descriptors in graph based MERC frameworks without significant loss in performance. In some cases, modest improvements were observed, particularly on IEMOCAP where the target speaker could be isolated.

The choice of text encoder affects performance depending on the length of VLM-generated descriptions. While Longformer can handle sequences up to 4096 tokens, most descriptions were within the 512-token limit of BERT. Consequently, BERT encoded the majority of samples efficiently without excessive padding, yielding slightly better performance than Longformer.

Dataset characteristics also influence the effectiveness of semantic visual features. IEMOCAP clips have relatively low spatial resolution (360×480 after speaker-focused splitting) and short durations, limiting the visibility of fine-grained facial expressions and the temporal context available for emotional inference. Additionally, VLM outputs occasionally contain hallucinated or repetitive descriptions, introducing noise into the semantic embeddings.

On MELD, multi-party interactions complicate speaker-specific emotion recognition. Multiple characters appearing simultaneously make it difficult for the VLM to isolate the target speaker, often resulting in descriptions that focus on overall scene context rather than the emotional state of the intended individual. Hallucinations and inconsistent outputs, sometimes exacerbated by larger batch sizes during generation, add further noise to the embeddings.

Frame-level analysis with BLIP-2 [20], where three representative frames from each clip (beginning, middle, end) were used, yielded more focused descriptions centered on the speaker's emotional state. This approach reduced redundancy and improved performance slightly compared to full-scene VLM descriptions, suggesting that targeted visual reasoning on key frames can be advantageous, particularly in multi-party settings.

Overall, these observations highlight that semantic visual representations are a viable alternative to handcrafted features in MERC, but their effectiveness is highly dependent on scene structure, speaker isolation, and quality of VLM outputs. Future improvements may be achieved through explicit speaker localization, noise reduction in VLM descriptions, and optimized frame sampling strategies for temporal emotion tracking.

6 Conclusion

This work explores an alternative approach to representing visual information in graph based Multimodal Emotion Recognition in Conversations (MERC). Instead of relying on conventional pixel-level visual embeddings, we employ a Vision–Language Model (VLM) to transform utterance-level video segments into structured semantic descriptions that capture observable behavioral cues and their temporal progression.

These semantically grounded representations provide a different perspective on incorporating visual signals within conversational reasoning frameworks and demonstrate that visual information can be expressed in a form closer to how emotions are typically interpreted. Our findings suggest that such semantic visual representations can serve as a viable alternative to traditional visual feature pipelines used in graph based MERC systems.

Future work may extend this idea to the acoustic modality by transforming speech signals into descriptive semantic representations that capture emotionally relevant characteristics of the audio. Additionally, architectures specifically designed to process semantic multimodal representations could further improve how such information is incorporated into conversational emotion recognition systems.

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